

Third SEMESTER-B.TECH- Course Work
END-SEMESTER EXAMINATION, Nov-Dec. 2024

Course Code-ICMTC301/EEMTC301/CMMTC301

Course Title-Engineering Mathematics

Time: 3:00 hours

Max Marks-50

Note: Attempt all questions. Missing data/information (if any), may be suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO																
	Attempt any two parts of the following																		
1 (a)	Expand the function $f(x) = x \sin x$ as a Fourier series in the interval $-\pi \leq x \leq \pi$. Hence deduce $\frac{1}{1.3} - \frac{1}{3.5} + \frac{1}{5.7} - \frac{1}{7.9} + \dots = \frac{\pi-2}{4}$.	5.0	CO1																
1(b)	Obtain the half range sine series for the function $f(x) = x^2$ in the interval $0 < x < 3$.	5.0	CO1																
1 (c)	Find the harmonics a_0, a_1, b_1 for the function $f(x)$ given by the following table	5.0	CO1																
	<table border="1" style="margin-left: auto; margin-right: auto;"> <tr> <td>x°:</td> <td>0</td> <td>60</td> <td>120</td> <td>180</td> <td>240</td> <td>300</td> <td>360</td> </tr> <tr> <td>$f(x)$:</td> <td>0.8</td> <td>0.6</td> <td>0.4</td> <td>0.7</td> <td>0.9</td> <td>1.1</td> <td>0.8</td> </tr> </table>	x° :	0	60	120	180	240	300	360	$f(x)$:	0.8	0.6	0.4	0.7	0.9	1.1	0.8		
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$f(x)$:	0.8	0.6	0.4	0.7	0.9	1.1	0.8												
	Attempt any two parts of the following																		
2 (a)	Verify Green's theorem in the plane for $\int_C [(x^2 - xy^3)dx + (y^2 - 2xy)dy]$, where C is the square with vertices $(0, 0), (2, 0), (2, 2), (0, 2)$.	5.0	CO2																
2 (b)	Verify divergence theorem for $\vec{F} = (x^2 - yz)\hat{i} + (y^2 - xz)\hat{j} + (z^2 - xy)\hat{k}$ taken over the rectangle parallelepiped $0 \leq x \leq a, 0 \leq y \leq b, 0 \leq z \leq c$.	5.0	CO2																
2(c)	If $\phi(x, y) = \log_e \sqrt{x^2 + y^2}$, show that $\text{grad } \phi = \frac{\vec{r} - (\hat{k} \cdot \vec{r})\hat{k}}{(\vec{r} - (\hat{k} \cdot \vec{r})\hat{k}) \cdot (\vec{r} - (\hat{k} \cdot \vec{r})\hat{k})}$.	5.0	CO2																
	Attempt any two parts of the following																		
3 (a)	Express the function $F(t) = 2t$, when $0 \leq t \leq 5$ and $F(t) = 10$ when $t > 5$ in terms of unit step function and hence obtain its Laplace transform.	5.0	CO3																

Q. No.	Question	Marks	CO
3(b)	Use Laplace transform to solve $\frac{d^2y}{dt^2} + a\frac{dy}{dt} - 2a^2y = 0$ given $y(0) = 6, y'(0) = 0$.	5.0	CO3
3 (c)	For the periodic function $f(t)$ of period 4 defined by $f(t) = 3t$, if $0 < t < 2$ and $f(t) = 6$, if $2 < t < 4$ find $L(f(t))$.	5.0	CO3
	Attempt any two parts of the following		
4 (a)	Show that when n is positive integer $P_n(x) = \frac{1}{n!2^n} \frac{d^n}{dx^n} (x^2 - 1)^n$.	5.0	CO4
4 (b)	Show that for Legendre's polynomial $P_n(x)$, we have $nP_n(x) = xP_n'(x) - P_{n-1}'(x)$.	5.0	CO4
4 (c)	Establish the orthogonality property of $P_n(x)$. $\int_{-1}^1 P_m(x)P_n(x)dx = 0, m \neq n$ and $\int_{-1}^1 [P_n(x)]^2 dx = \frac{2}{2n+1}$.	5.0	CO4
	Attempt any two parts of the following		
5 (a)	Solve $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ in the interval $0 \leq x \leq \pi$, subject to conditions: $u(0, y) = 0, u(\pi, y) = 0, u(x, 0) = 1$ and $u(x, y) \rightarrow 0$ as $y \rightarrow \infty$ for all x .	5.0	CO2
5 (b)	Solve the partial differential equation $\frac{\partial^2 y}{\partial t^2} = a^2 \frac{\partial^2 y}{\partial x^2}$ representing the vibration of a string of length l , fixed at both ends, subject to boundary conditions $y(0, t) = 0, y(l, t) = 0$ and initial conditions $y = y_0 \sin \frac{\pi x}{l}$ and $\frac{\partial y}{\partial t} = 0$ at $t = 0$.	5.0	CO2
5 (c)	Find the temperature in a metal rod of length l with both ends insulated and with initial temperature in the rod $\sin \frac{\pi x}{l}$.	5.0	CO2